

7 (a) Describe the structure of a nuclear atom.

.....

.....

..... [2]

(b) P and Q are radioactive nuclei. P has 94 protons. Q has 91 protons.

Each nucleus decays to form a nucleus with 92 protons.

(i) State the type of radioactive decay experienced by P.

..... [1]

(ii) State the type of radioactive decay experienced by Q.

..... [1]

(c) Particle X has a ratio of $\frac{\text{charge}}{\text{mass}}$ equal to $+1.76 \times 10^{11} \text{ C kg}^{-1}$.

Identify particle X. Show your working.

particle X [2]

[Total: 6]